

N 697

Seat No.

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2025 VII 02 1100 – N 697– MATHEMATICS (71) GEOMETRY—PART II (E)
(REVISED COURSE)

Time : 2 Hours

(Pages 11)

Max. Marks : 40

Note :—

- (i) All questions are compulsory.
- (ii) Use of calculator is not allowed.
- (iii) The numbers to the right of the questions indicate full marks.
- (iv) In case of MCQs [Q. No. 1(A)] only the first attempt will be evaluated and will be given credit.
- (v) Draw proper figures wherever necessary.
- (vi) The marks of construction should be clear. Do not erase them.
- (vii) Diagram is essential for writing the proof of the theorem.

1. (A) Choose the correct alternative from given :

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(1) If $\triangle ABC \sim \triangle DEF$, $m\angle B = 60^\circ$, then $m\angle E = \dots\dots\dots$

- (A) 30°
- (B) 60°
- (C) 90°
- (D) 45°

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- (2) Two circles of radii 5.5 cm and 4.2 cm touch each other externally then distance between their centres is
- (A) 9.7 cm
- (B) 1.3 cm
- (C) 5.5 cm
- (D) 4.2 cm
- (3) A line makes an angle of 45° with the positive direction of X-axis. So the slope of the line is
- (A) $\frac{1}{2}$
- (B) $\frac{\sqrt{3}}{2}$
- (C) 1
- (D) $\sqrt{3}$
- (4) The volume of a cube of side 2 cm is
- (A) 4 cm^3
- (B) 2 cm^3
- (C) 6 cm^3
- (D) 8 cm^3

(B) Solve the following subquestions :

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- (1) Find the diagonal of square whose side is 10 cm.
- (2) The ratio of corresponding sides of similar triangles is 3 : 5, then find the ratio of their areas.

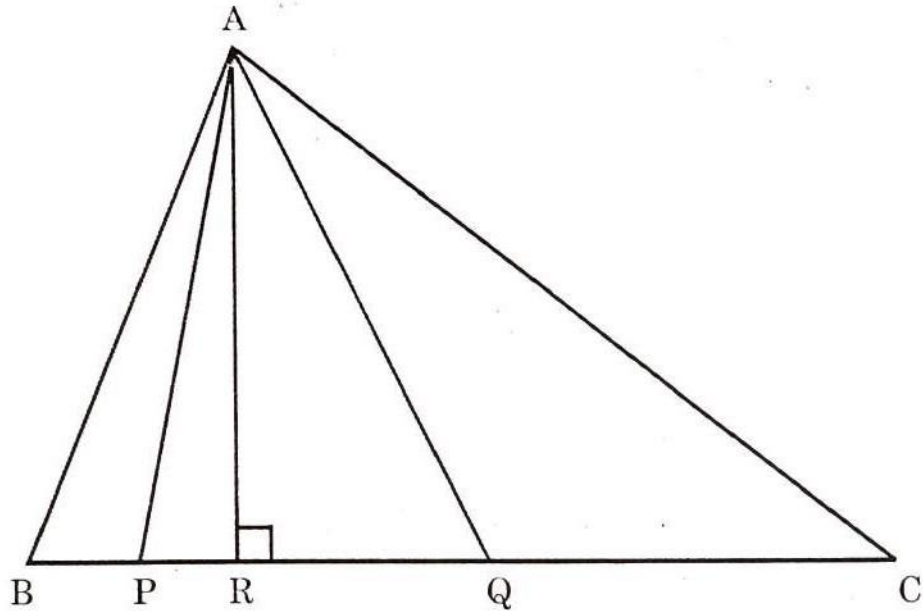
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(3) Find the slope of the line passing through the points A(2, 3) and B(4, 7).

(4) If $\sin \theta = \frac{7}{25}$, then find the value of $\operatorname{cosec} \theta$.

2. (A) Complete the following activities and rewrite it (any two) : 4

(1) In the given figure, $AR \perp BC$, $AR \perp PQ$, then complete the activity for finding $\frac{A(\triangle ABC)}{A(\triangle APQ)}$.



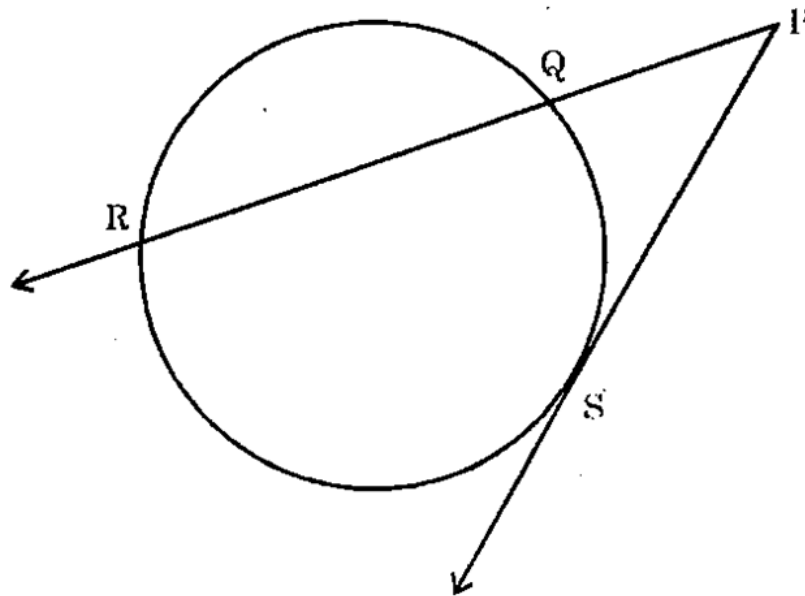
Activity :

$$\frac{A(\triangle ABC)}{A(\triangle APQ)} = \frac{\boxed{} \times AR}{PQ \times \boxed{}}$$

$$\therefore \frac{A(\triangle ABC)}{A(\triangle APQ)} = \frac{\boxed{}}{\boxed{}}$$

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- (2) In the following figure, seg PS is a tangent segment, line PR is a secant. If $PQ = 3.6$, $QR = 6.4$, then find PS by completing the following activity.



Activity :

$$PS^2 = PQ \times \underline{\hspace{1cm}}$$

(tangent secant segments theorem)

$$\therefore PS^2 = PQ \times (PQ + \underline{\hspace{1cm}})$$

$$\therefore PS^2 = 3.6 \times (3.6 + \underline{\hspace{1cm}})$$

$$\therefore PS^2 = 3.6 \times 10$$

$$\therefore PS^2 = 36$$

$$\therefore PS = \underline{\hspace{1cm}}$$

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- (3) Measure of an arc of a circle is 90° and its radius is 14 cm. Complete the following activity to find the length of an arc.

Activity :

$$\begin{aligned}\text{Length of an arc} &= \frac{\theta}{360} \times \boxed{} \dots\dots (\text{Formula}) \\ &= \frac{90}{360} \times 2 \times \frac{22}{7} \times \boxed{} \\ &= \frac{1}{4} \times \boxed{}\end{aligned}$$

$$\text{Length of an arc} = \boxed{} \text{ cm}$$

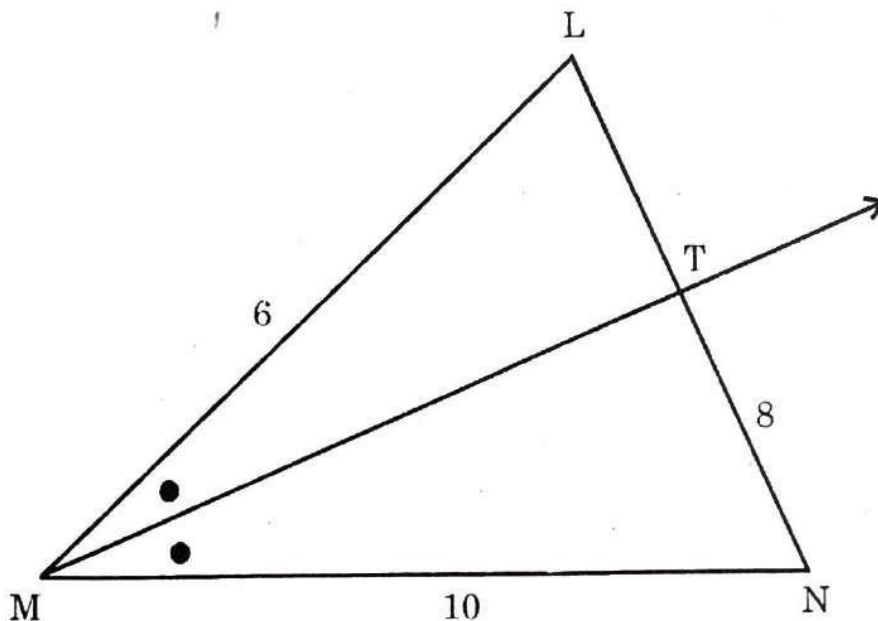
(B) Solve the following subquestions (any four) :

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- (1) In the following figure

In $\triangle LMN$, ray MT bisects $\angle LMN$.

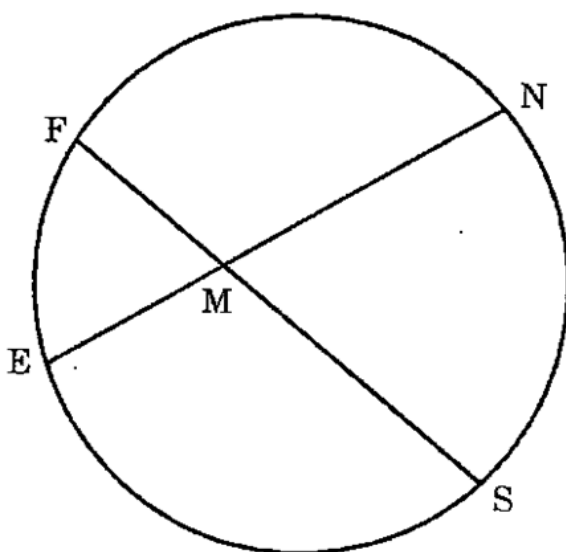
If $LM = 6$, $MN = 10$, $TN = 8$, then find LT .



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- (2) Find surface area of sphere of radius 7 cm.
- (3) In the following figure, $m(\text{arc NS}) = 125^\circ$, $m(\text{arc EF}) = 37^\circ$. Find $m\angle NMS$.



- (4) Find the co-ordinates of midpoint of the segment joining the points $P(22, 20)$ and $Q(0, 16)$.
- (5) Find the volume of a cone if radius of its base is 7 cm and its perpendicular height is 15 cm.

3. (A) Complete the following activities and rewrite it (any one) : 3

- (1) If $\tan \theta = 1$, then find the value of $\frac{\sin \theta + \cos \theta}{\sec \theta + \operatorname{cosec} \theta}$ by completing the following activity.

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Activity :

$$\tan \theta = 1 \text{ (given)}$$

$$\text{but } \tan \boxed{} = 1$$

$$\therefore \theta = \boxed{}$$

$$\therefore \frac{\sin \theta + \cos \theta}{\sec \theta + \operatorname{cosec} \theta} = \frac{\sin 45^\circ + \cos 45^\circ}{\sec 45^\circ + \operatorname{cosec} 45^\circ}$$

$$= \frac{\frac{1}{\boxed{}} + \frac{1}{\sqrt{2}}}{\sqrt{2} + \boxed{}}$$

$$= \frac{\frac{2}{\sqrt{2}}}{\boxed{}}$$

$$\frac{\sin \theta + \cos \theta}{\sec \theta + \operatorname{cosec} \theta} = \frac{1}{\boxed{}}$$

- (2) In the following figure, point O is the centre of the circle and length of chord AB is equal to the radius of the circle. Find the measures of :

(i) $\angle AOB$

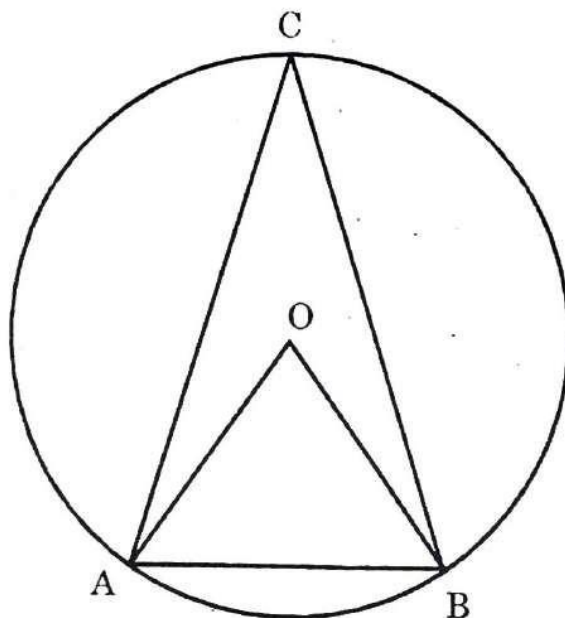
(ii) arc AB

(iii) $\angle ACB$

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by completing the activity.



Activity :

In $\triangle AOB$,

$$AO = OB = AB$$

$\therefore \triangle AOB$ is an triangle.

$$\therefore m\angle AOB = \text{$$

$$m\angle AOB = m(\text{arc } AB) = \text{} \quad (\text{definition of measure of an arc})$$

$$m\angle ACB = \frac{1}{2} \times \text{$$

.....

$$= \frac{1}{2} \times 60^\circ$$

$$m\angle ACB = \text{$$

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(B) Solve the following subquestions (any two) :

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- (1) Find co-ordinates of point P, if P divides the line segment joining the points A(-1, 7) and B(4, -3) in the ratio 2 : 3.
- (2) Draw a circle with centre O of radius 3.4 cm. Draw a chord MN of length 5.7 cm in it. Construct tangents at point M and N to the circle.
- (3) A storm broke a tree and the treetop rested 20 m from the base of the tree, making an angle of 60° with the horizontal. Find the height of the tree.
- (4) Prove that, 'In a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of remaining two sides'.

4. Solve the following subquestions (any two) :

8

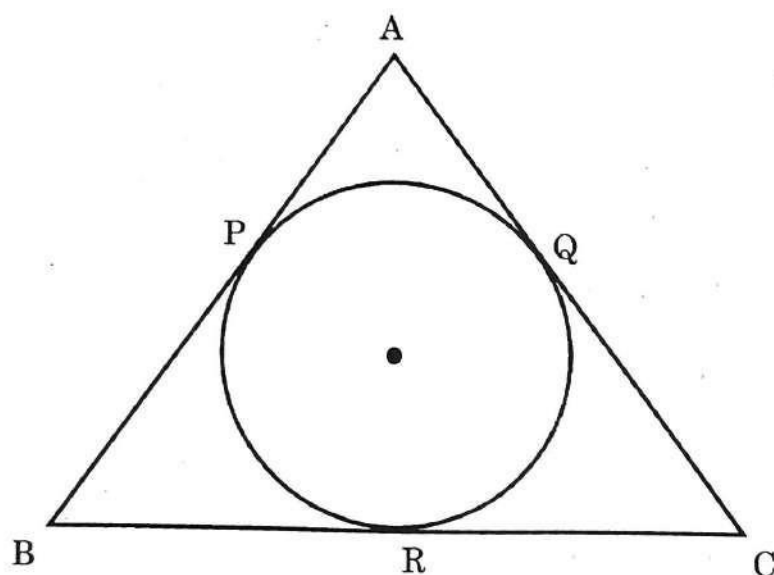
- (1) $\triangle ABC$ has sides of length 4 cm, 5 cm and 6 cm while $\triangle PQR$ has perimeter of 90 cm. If $\triangle ABC$ is similar to $\triangle PQR$, then find the length of corresponding sides of $\triangle PQR$.
- (2) $\triangle ABC \sim \triangle PBR$, $BC = 8$ cm, $AC = 10$ cm, $\angle B = 90^\circ$, $\frac{BC}{BR} = \frac{5}{4}$, then construct $\triangle PBR$.

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- (3) In the following figure $\triangle ABC$ is an isosceles triangle with perimeter 44 cm. The base BC is of length 12 cm. Side AB and AC are congruent. A circle touches the three sides of triangle as shown.

Find the length of tangent segment from A to circle.



5. Solve the following subquestions (any one) :

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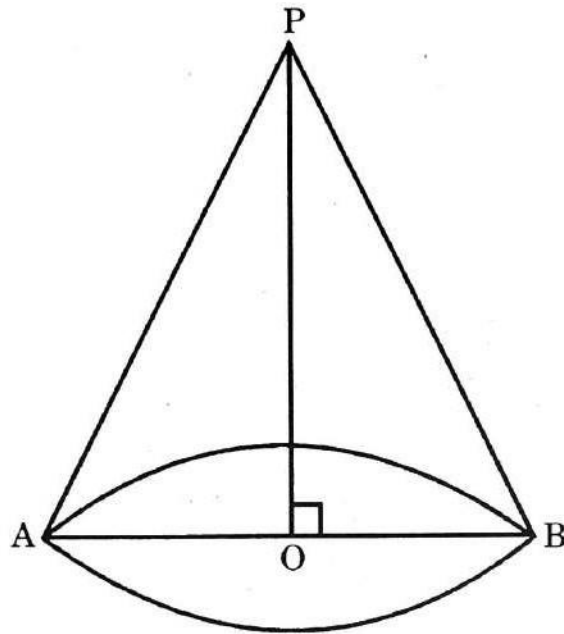
- (1) Draw right-angled $\triangle ABC$ of lengths of sides are 3 cm, 4 cm and 5 cm. Draw median on the hypotenuse of $\triangle ABC$.

Then :

- (i) Measure the length of median and write it.
- (ii) By observing lengths of median and hypotenuse write your observations.

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- (2) Observe the given figure and answer the following questions :



- (i) How many surfaces does a solid cone have ?
- (ii) What are the names of slant height and perpendicular height in the given figure ?
- (iii) If slant height of solid cone is 10 cm and perpendicular height is 8 cm, then find diameter of base of solid cone ?